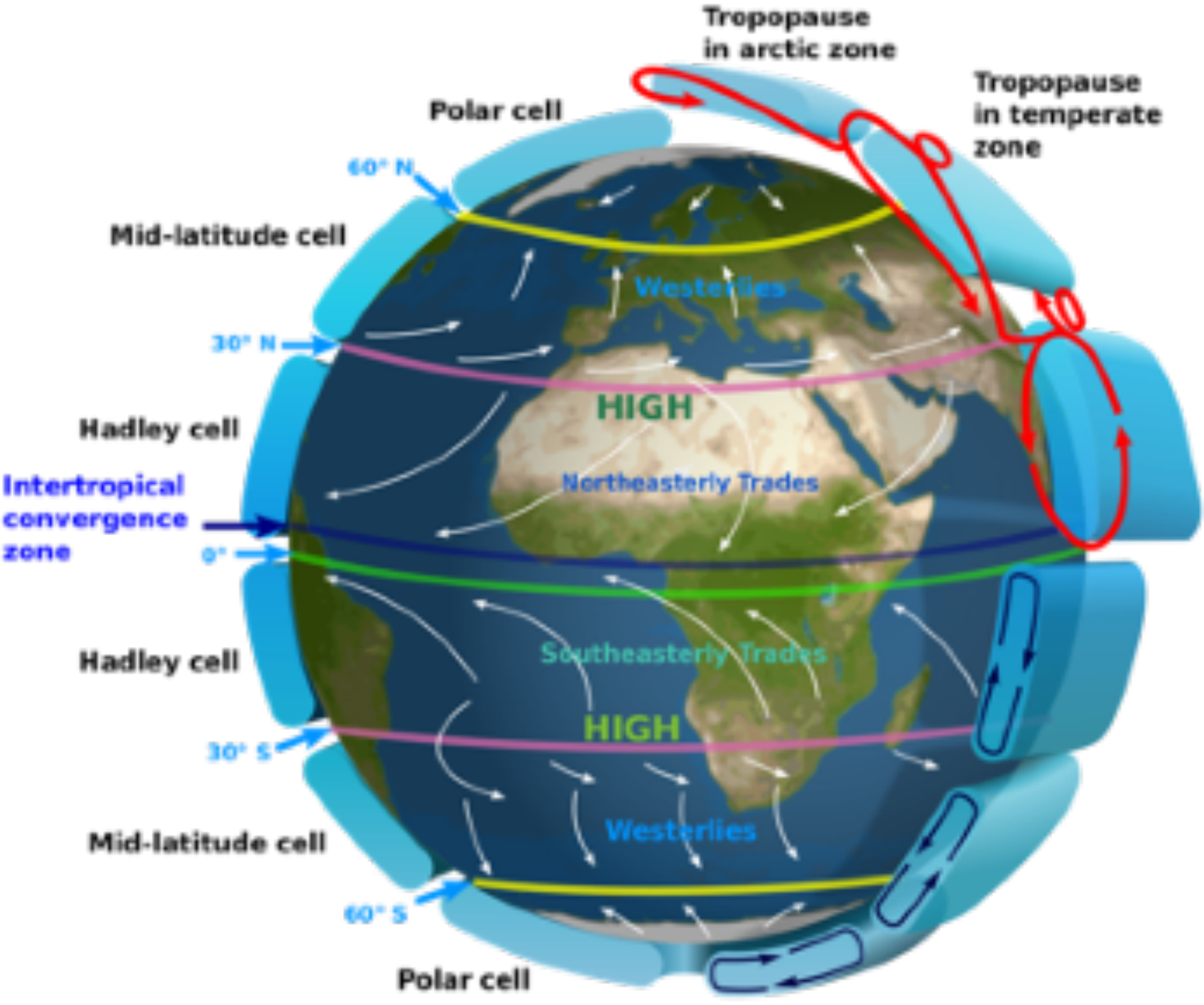
Appendix: region(s) of focus at MPI-M

34

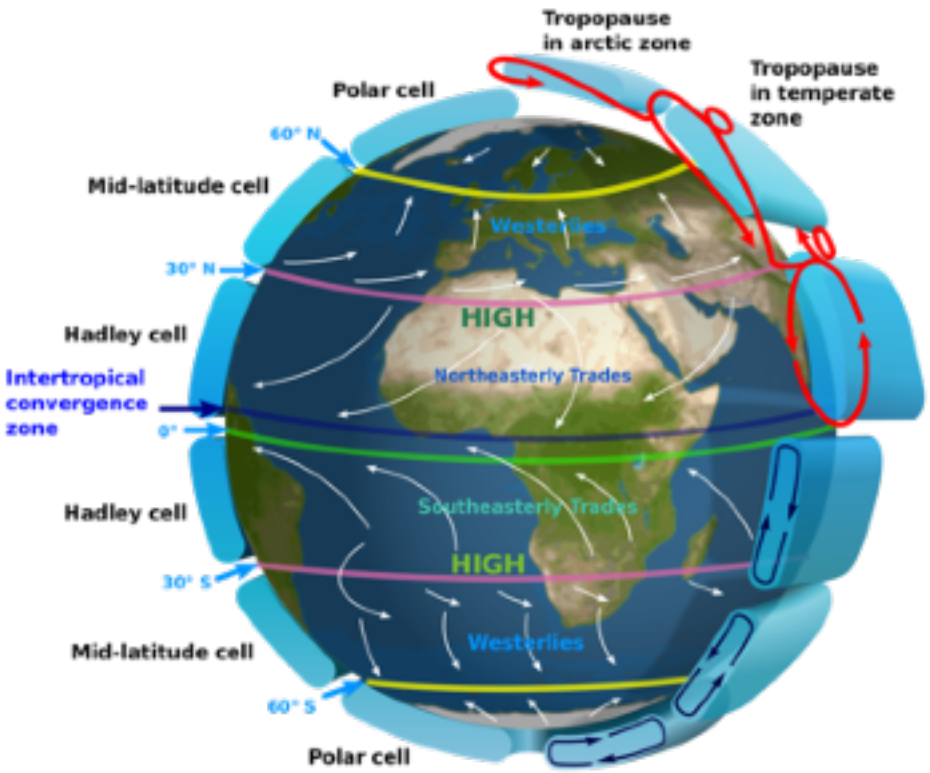

Tropopause
in arctic zone
Tropopause
in temperate
zone
Polar cell
60° N
Mid-latitude cell
Westerlies
30° N
HIGH
Hadley cell
Northeasterly Trades
Intertropical
convergence
zone
0°
Hadley cell
Southeasterly Trades
HIGH
30° S
Mid-latitude cell
Westerlies
60° S
Polar cell

- Trade wind region:
  - low-level clouds reflect incoming SW radiation and trap outgoing LW radiation
  - trade cumulus is widespread

- Intertropical convergence zone (ITCZ):
  - deep convection and high cloud cover
  - controls monsoons
  - produces rainfall in tropical forests
  - drives large-scale dynamics

Appendix: region(s) of focus at MPI-M

- Trade wind region:
  - low-level clouds reflect incoming SW radiation and trap outgoing LW radiation
  - trade cumulus is widespread
- Intertropical convergence zone (ITCZ):
  - deep convection and high cloud cover
  - controls monsoons
  - produces rainfall in tropical forests
  - drives large-scale dynamics



Appendix: serializing campaigns over many years allows for climatological analysis

- Global Atmospheric Research Program (GARP) Atlantic Tropical Experiment (GATE)
- 50+ years of measurements in the tropical Atlantic!

courtesy of UCAR